

Specialized Bicycle Components was interested in developing a new front suspension fork for the mountain bike market. Although the firm was already selling suspension forks on its bicycles (Exhibit 6-1), it was interested in exploring designs that would provide higher value for the recreational cyclist.

The development team had spent a great deal of time identifying customer needs. In addition to logging many hours of riding on suspended bikes themselves, the members of the team had interviewed lead users at mountain bike races and recreational cyclists on local trails, and they also had spent time working with dealers in their stores. As a result of this process they had assembled a list of customer needs. They now faced several challenges:

- How could the relatively subjective customer needs be translated into precise targets for the remaining development effort?
- How could the team and its senior management agree on what would constitute success or failure of the resulting product design?
- How could the team develop confidence that its intended product would garner a substantial share of the suspension fork market?
- How could the team resolve the inevitable trade-offs among product characteristics like cost and weight?

This chapter presents a method for establishing product specifications. We assume that the customer needs are already documented as described in Chapter 5, Identifying Customer Needs. The method employs several simple information systems, all of which can be constructed using conventional spreadsheet software.

What Are Specifications?

Customer needs are generally expressed in the “language of the customer.” The primary customer needs for the suspension fork are listed in Exhibit 6-2. Customer needs such as “the suspension is easy to install” or “the suspension enables high-speed descents on bumpy trails” are typical in terms of the subjective quality of the expressions; however, while such expressions are helpful in developing a clear sense of the issues of interest to customers, they provide little specific guidance about how to design and engineer the product. They simply leave too much margin for subjective interpretation. For this reason, development teams usually establish a set of specifications, which spell out in precise, measurable detail *what* the product has to do. Product specifications do not tell the team *how* to address the customer needs, but they do represent an unambiguous agreement on what the team will attempt to achieve to satisfy the customer needs. For example, in contrast to the customer need that “the suspension is easy to install,” the corresponding specification might be that “the average time to assemble the fork to the frame is less than 75 seconds.”

We intend the term *product specifications* to mean the precise description of what the product has to do. Some firms use the terms “product requirements” or “engineering characteristics” in this way. Other firms use “specifications” or “technical specifications” to refer to key design variables of the product such as the oil viscosity or spring constant of the suspension system. These are just differences in terminology. For clarity, let us be

EXHIBIT 6-2

Customer needs for the suspension fork and their relative importance (shown in a convenient spreadsheet format).

No.		Need	Imp.
1	The suspension	reduces vibration to the hands.	3
2	The suspension	allows easy traversal of slow, difficult terrain.	2
3	The suspension	enables high-speed descents on bumpy trails.	5
4	The suspension	allows sensitivity adjustment.	3
5	The suspension	preserves the steering characteristics of the bike.	4
6	The suspension	remains rigid during hard cornering.	4
7	The suspension	is lightweight.	4
8	The suspension	provides stiff mounting points for the brakes.	2
9	The suspension	fits a wide variety of bikes, wheels, and tires.	5
10	The suspension	is easy to install.	1
11	The suspension	works with fenders.	1
12	The suspension	instills pride.	5
13	The suspension	is affordable for an amateur enthusiast.	5
14	The suspension	is not contaminated by water.	5
15	The suspension	is not contaminated by grunge.	5
16	The suspension	can be easily accessed for maintenance.	3
17	The suspension	allows easy replacement of worn parts.	1
18	The suspension	can be maintained with readily available tools.	3
19	The suspension	lasts a long time.	5
20	The suspension	is safe in a crash.	5

precise about a few definitions. A *specification* (singular) consists of a *metric* and a *value*. For example, “average time to assemble” is a metric, while “less than 75 seconds” is the value of this metric. Note that the value may take on several forms, including a particular number, a range, or an inequality. Values are always labeled with the appropriate units (e.g., seconds, kilograms, joules). Together, the metric and value form a specification. The *product specifications* (plural) are simply the set of the individual specifications.

When Are Specifications Established?

In an ideal world, the team would establish the product specifications once early in the development process and then proceed to design and engineer the product to exactly meet those specifications. For some products, such as soap or soup, this approach works quite well; the technologists on the team can reliably concoct a formulation that satisfies almost any reasonable specifications; however, for technology-intensive products this is rarely possible. For such products, specifications are established at least twice. Immediately after identifying the customer needs, the team sets *target specifications*. These specifications represent the hopes and aspirations of the team, but they are established before the team knows what constraints the product technology will place on what

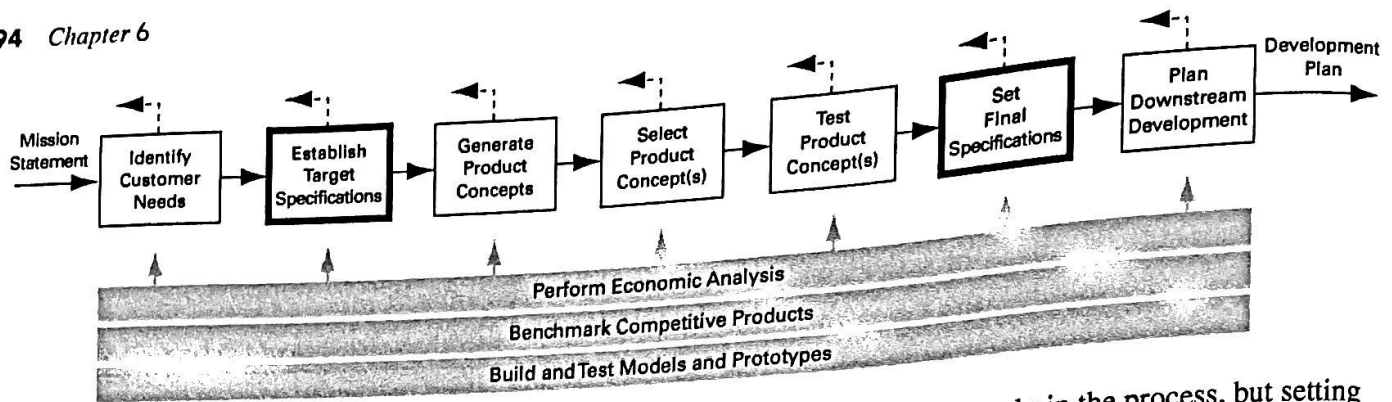


EXHIBIT 6-3 The concept development process. The target specifications are set early in the process, but setting the final specifications must wait until after the product concept has been selected.

can be achieved. The team's efforts may fail to meet some of these specifications and may exceed others, depending on the product concept the team eventually selects. For this reason, the target specifications must be refined after a product concept has been selected. The team revisits the specifications while assessing the actual technological constraints and the expected production costs. To set the *final specifications*, the team must frequently make hard trade-offs among different desirable characteristics of the product. For simplicity, we present a two-stage process for establishing specifications, but we note that in some organizations specifications are revisited many times throughout the development process.

The two stages in which specifications are established are shown as part of the concept development process in Exhibit 6-3. Note that the final specifications are one of the key elements of the development plan, which is usually documented in the project's *contract book*. The contract book (described in Chapter 19, Managing Projects) specifies what the team agrees to achieve, the project schedule, the required resources, and the economic implications for the business. The list of product specifications is also one of the key information systems used by the team throughout the development process.

This chapter presents two methods: the first is for establishing the target specifications and the second is for setting the final specifications after the product concept has been selected.

Establishing Target Specifications

As Exhibit 6-3 illustrates, the target specifications are established after the customer needs have been identified but before product concepts have been generated and the most promising one(s) selected. An arbitrary setting of the specifications may not be technically feasible. For example, in designing a suspension fork, the team cannot assume in advance that it will be able to achieve simultaneously a mass of 1 kilogram, a manufacturing cost of \$30, and the best descent time on the test track, as these are three quite aggressive specifications. Actually meeting the specifications established at this point is contingent upon the details of the product concept the team eventually selects. For this reason, such preliminary specifications are labeled "target specifications." They are the goals of the development team, describing a product that the team believes would succeed in the marketplace. Later these specifications will be refined based on the limitations of the product concept actually selected.

The process of establishing the target specifications contains four steps:

1. Prepare the list of metrics.
2. Collect competitive benchmarking information.
3. Set ideal and marginally acceptable target values.
4. Reflect on the results and the process.

Step 1: Prepare the List of Metrics

The most useful metrics are those that reflect as directly as possible the degree to which the product satisfies the customer needs. The relationship between needs and metrics is central to the entire concept of specifications. The working assumption is that a translation from customer needs to a set of precise, measurable specifications is possible and that meeting specifications will therefore lead to satisfaction of the associated customer needs.

A list of metrics is shown in Exhibit 6-4. A good way to generate the list of metrics is to contemplate each need in turn and to consider what precise, measurable characteristic of the product will reflect the degree to which the product satisfies that need. In the ideal case, there is one and only one metric for each need. In practice, this is frequently not possible.

For example, consider the need that the suspension be "easy to install." The team may conclude that this need is largely captured by measuring the time required for assembly of the fork to the frame; however, note the possible subtleties in this translation. Is assembly time really identical to ease of installation? The installation could be extremely fast but require an awkward and painful set of finger actions, which ultimately may lead to worker injury or dealer frustration. Because of the imprecise nature of the translation process, those establishing the specifications should have been directly involved in identifying the customer needs. In this way the team can rely on its understanding of the meaning of each need statement derived from firsthand interactions with customers.

The need for the fork to reduce vibration to the user's hands may be even more difficult to translate into a single metric, because there are many different conditions under which vibration can be transmitted, including small bumps on level roads and big bumps on rough trails. The team may conclude that several metrics are required to capture this need, including, for example, the metrics "attenuation from dropout to handlebar at 10 Hz" and "maximum value from the Monster." (The "Monster" is a suspension test developed by *Mountain Bike* magazine.)

A simple needs-metrics matrix represents the relationship between needs and metrics. An example needs-metrics matrix is shown in Exhibit 6-5. The rows of the matrix correspond to the customer needs, and the columns of the matrix correspond to the metrics. A mark in a cell of the matrix means that the need and the metric associated with the cell are related; performance relative to the metric will influence the degree to which the product satisfies the customer need. This matrix is a key element of the *House of Quality*, a graphical technique used in *Quality Function Deployment*, or *QFD* (Hauser and Clausing, 1988). In many cases, we find the information in the needs-metrics matrix is just as easily communicated by listing the numbers of the needs related to each metric alongside the list of metrics (the second column in Exhibit 6-4). There are some cases, however, in which the mapping from needs to metrics is complex, and the matrix can be quite useful for representing this mapping.

Metric No.	Need Nos.	Metric	Imp.	Units
1	1, 3	Attenuation from dropout to handlebar at 10 Hz	3	dB
2	2, 6	Spring preload	3	N
3	1, 3	Maximum value from the Monster	5	g
4	1, 3	Minimum descent time on test track	5	s
5	4	Damping coefficient adjustment range	3	N-s/m
6	5	Maximum travel (26-in. wheel)	3	mm
7	5	Rake offset	3	mm
8	6	Lateral stiffness at the tip	3	kN/m
9	7	Total mass	4	kg
10	8	Lateral stiffness at brake pivots	2	kN/m
11	9	Headset sizes	5	in.
12	9	Steertube length	5	mm
13	9	Wheel sizes	5	List
14	9	Maximum tire width	5	in.
15	10	Time to assemble to frame	1	s
16	11	Fender compatibility	1	List
17	12	Instills pride	5	Subj.
18	13	Unit manufacturing cost	5	US\$
19	14	Time in spray chamber without water entry	5	s
20	15	Cycles in mud chamber without contamination	5	k-cycles
21	16, 17	Time to disassemble/assemble for maintenance	3	s
22	17, 18	Special tools required for maintenance	3	List
23	19	UV test duration to degrade rubber parts	5	hr
24	19	Monster cycles to failure	5	Cycles
25	20	Japan Industrial Standards test	5	Binary
26	20	Bending strength (frontal loading)	5	kN

EXHIBIT 6-4 List of metrics for the suspension. The relative importance of each metric and the units for the metric are also shown. “Subj.” is an abbreviation indicating that a metric is subjective.

- A few guidelines should be considered when constructing the list of metrics:
- **Metrics should be complete.** Ideally each customer need would correspond to a single metric, and the value of that metric would correlate perfectly with satisfaction of that need. In practice, several metrics may be necessary to completely reflect a single customer need.
 - **Metrics should be dependent, not independent, variables.** This guideline is a variant of the *what-not-how principle* introduced in Chapter 5. As do customer needs, specifications also indicate *what* the product must do, but not *how* the specifications will

Need	Metric																									
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
1	Reduces vibration to the hands	●																								
2	Allows easy traversal of slow, difficult terrain		●																							
3	Enables high-speed descents on bumpy trails	●		●																						
4	Allows sensitivity adjustment				●																					
5	Preserves the steering characteristics of the bike					●	●																			
6	Remains rigid during hard cornering	●						●																		
7	Is lightweight								●																	
8	Provides stiff mounting points for the brakes									●																
9	Fits a wide variety of bikes, wheels, and tires										●	●	●	●												
10	Is easy to install														●											
11	Works with fenders															●										
12	Instills pride																●									
13	Is affordable for an amateur enthusiast																	●								
14	Is not contaminated by water																		●							
15	Is not contaminated by grunge																			●						
16	Can be easily accessed for maintenance																				●					
17	Allows easy replacement of worn parts																				●	●				
18	Can be maintained with readily available tools																					●				
19	Lasts a long time																						●	●		
20	Is safe in a crash																								●	●

EXHIBIT 6-5 The needs-metrics matrix.

be achieved. Designers use many types of variables in product development; some are *dependent*, such as the mass of the fork, and some are *independent*, such as the material used for the fork. In other words, designers cannot control mass directly because it arises from other independent decisions the designers will make, such as dimensions and materials choices. Metrics specify the overall performance of a product and should therefore be the dependent variables (i.e., the performance measures or output variables) in the design problem. By using dependent variables for the specifications, designers are left with the freedom to achieve the specifications using the best approach possible.

- ***Metrics should be practical.*** It does not serve the team to devise a metric for a bicycle suspension that can only be measured by a scientific laboratory at a cost of \$100,000. Ideally, metrics will be directly observable or analyzable properties of the product that can be easily evaluated by the team.
- ***Some needs cannot easily be translated into quantifiable metrics.*** The need that the suspension instills pride may be quite critical to success in the fashion-conscious mountain bike market, but how can pride be quantified? In these cases, the team simply repeats the need statement as a specification and notes that the metric is subjective. (We indicate this by entering "Subj." in the units column.) However, the team still has to decide how to confirm that the specification is satisfied; for example, by a panel of customers.
- ***The metrics should include the popular criteria for comparison in the marketplace.*** Many customers in various markets buy products based on independently published evaluations. Such evaluations are found, for example, in *Popular Science*, *Consumer Reports*, on various Internet sites, or, in our case, in *Bicycling* and *Mountain Bike* magazines. If the team knows that its product will be evaluated by the trade media and knows what the evaluation criteria will be, then it should include metrics corresponding to these criteria. *Mountain Bike* magazine uses a test machine called the Monster, which measures the vertical acceleration (in g's) of the handlebars as a bicycle equipped with the fork runs over a block 50 millimeters tall. For this reason, the team included "maximum value from the Monster" as a metric. If the team cannot find a relationship between the criteria used by the media and the customer needs it has identified, then it should ensure that a need has not been overlooked and/or should work with the media to revise the criteria. In a few cases, the team may conclude that high performance in the media evaluations is in itself a customer need and choose to include a metric used by the media that has little intrinsic technical merit.

In addition to denoting the needs related to each metric, Exhibit 6-4 contains the units of measurement and an importance rating for each metric. The units of measurement are most commonly conventional engineering units such as kilograms and seconds; however, some metrics will not lend themselves to numerical values. The need that the suspension "works with fenders" is best translated into a specification listing the models of fenders with which the fork is compatible. In this case, the value of the metric is actually a list of fenders rather than a number. For the metric involving the standard safety test, the value is pass/fail. (We indicate these two cases by entering "List" and "Binary" in the units column.)

The importance rating of a metric is derived from the importance ratings of the needs it reflects. For cases in which a metric maps directly to a single need, the importance rating of the need becomes the importance rating of the metric. For cases in which a metric is related to more than one need, the importance of the metric is determined by considering the importances of the needs to which it relates and the nature of these relationships. We believe that there are enough subtleties in this process that importance weightings can best be determined through discussion among the team members, rather than through a formal algorithm. When there are relatively few specifications and establishing the relative importance of these specifications is critically important, *conjoint analysis* may be useful. Conjoint analysis is described briefly later in this chapter and publications explaining the technique are referenced at the end of the chapter.

Step 2: Collect Competitive Benchmarking Information

Unless the team expects to enjoy a total monopoly, the relationship of the new product to competitive products is paramount in determining commercial success. While the team will have entered the product development process with some idea of how it wishes to compete in the marketplace, the target specifications are the language the team uses to discuss and agree on the detailed positioning of its product relative to existing products, both its own and competitors'. Information on competing products must be gathered to support these positioning decisions.

An example of a competitive benchmarking chart is shown in Exhibit 6-6. The columns of the chart correspond to the competitive products and the rows are the metrics established in step 1. Note that the competitive benchmarking chart can be constructed as a simple appendage to the spreadsheet containing the list of metrics. (This information is one of the "rooms" in the *House of Quality*, described by Hauser and Clausing.)

The benchmarking chart is conceptually very simple. For each competitive product, the values of the metrics are simply entered down a column. Gathering these data can be very time consuming, involving (at the least) purchasing, testing, disassembling, and estimating the production costs of the most important competitive products; however, this investment of time is essential, as no product development team can expect to succeed without having this type of information. A word of warning: Sometimes the data contained in competitors' catalogs and supporting literature are not accurate. Where possible, values of the key metrics should be verified by independent testing or observation.

An alternative competitive benchmarking chart can be constructed with rows corresponding to the customer needs and columns corresponding to the competitive products (see Exhibit 6-7). This chart is used to compare customers' perceptions of the relative degree to which the products satisfy their needs. Constructing this chart requires collecting customer perception data, which can also be very expensive and time consuming. Some techniques for measuring customers' perceptions of satisfaction of needs are contained in a book by Urban and Hauser (1993). Both charts can be useful and any discrepancies between the two are instructive. At a minimum, a chart showing the competitive values of the metrics (Exhibit 6-6) should be created.

Step 3: Set Ideal and Marginally Acceptable Target Values

In this step, the team synthesizes the available information to actually set the *target values* for the metrics. Two types of target value are useful: an *ideal value* and a *marginally acceptable value*. The ideal value is the best result the team could hope for. The marginally acceptable value is the value of the metric that would just barely make the product commercially viable. Both of these targets are useful in guiding the subsequent stages of concept generation and concept selection, and for refining the specifications after the product concept has been selected.

There are five ways to express the values of the metrics:

- **At least X:** These specifications establish targets for the lower bound on a metric, but higher is still better. For example, the value of the brake mounting stiffness is specified to be at least 325 kilonewtons/meter.
- **At most X:** These specifications establish targets for the upper bound on a metric, with smaller values being better. For example, the value for the mass of the suspension fork is set to be at most 1.4 kilograms.

Metric No.	Need Nos.	Metric	Imp.	Units	ST Tritrack	Maniray 2	Rox Tahx Quadra	Rox Tahx TI 21	Tonka Pro	Gunhill Head Shox
1	1, 3	Attenuation from dropout to handlebar at 10 Hz	3	dB	8	15	10	15	9	13
2	2, 6	Spring preload	3	N	550	760	500	710	480	680
3	1, 3	Maximum value from the Monster	5	g	3.6	3.2	3.7	3.3	3.7	3.4
4	1, 3	Minimum descent time on test track	5	s	13	11.3	12.6	11.2	13.2	11
5	4	Damping coefficient adjustment range	3	N-s/m	0	0	0	200	0	0
6	5	Maximum travel (26-in. wheel)	3	mm	28	48	43	46	33	38
7	5	Rake offset	3	mm	41.5	39	38	38	43.2	39
8	6	Lateral stiffness at the tip	3	kN/m	59	110	85	85	65	130
9	7	Total mass	4	kg	1.409	1.385	1.409	1.364	1.222	1.100
10	8	Lateral stiffness at brake pivots	2	kN/m	295	550	425	425	325	650
11	9	Headset sizes	5	in.	1.000 1.125	1.000 1.125 1.250	1.000 1.125 1.250	1.000 1.125 1.250	1.000 1.125 1.250	NA
12	9	Steertube length	5	mm	150 180 210 230 255	140 165 190 215	150 170 190 210	150 170 190 210 230	150 190 210 220	NA
13	9	Wheel sizes	5	List	26 in.	26 in.	26 in.	26 in. 700C	26 in.	26 in.

EXHIBIT 6-6 Competitive benchmarking chart based on metrics.

- **Between X and Y:** These specifications establish both upper and lower bounds for the value of a metric. For example, the value for the spring preload is set to be between 480 and 800 newtons. Any more and the suspension is harsh; any less and the suspension is too bouncy.
- **Exactly X:** These specifications establish a target of a particular value of a metric, with any deviation degrading performance. For example, the ideal value for the rake offset metric is set to 38 millimeters. This type of specification is to be avoided if possible because such specifications substantially constrain the design. Often, upon reconsideration, the team realizes that what initially appears as an "exactly X" specification can be expressed as a "between X and Y" specification.
- **A set of discrete values:** Some metrics will have values corresponding to several discrete choices. For example, the headset diameters are 1.000, 1.125, or 1.250 inches. (Industry practice is to use English units for these and several other critical bicycle dimensions.)

Metric No.	Need Nos.	Metric	Imp.	Units	ST Tritrack	Maniray 2	Rox Tahx Quadra	Rox Tahx TI 21	Tonka Pro	Gunhill Head Shox
14	9	Maximum tire width	5	in.	1.5	1.75	1.5	1.75	1.5	1.5
15	10	Time to assemble to frame	1	s	35	35	45	45	35	85
16	11	Fender compatibility	1	List	Zefal	None	None	None	None	All
17	12	Instills pride	5	Subj.	1	4	3	5	3	5
18	13	Unit manufacturing cost	5	US\$	65	105	85	115	80	100
19	14	Time in spray chamber without water entry	5	s	1300	2900	>3600	>3600	2300	>3600
20	15	Cycles in mud chamber without contamination	5	k-cycles	15	19	15	25	18	35
21	16, 17	Time to disassemble/assemble for maintenance	3	s	160	245	215	245	200	425
22	17, 18	Special tools required for maintenance	3	List	Hex	Hex	Hex	Hex	Long hex	Hex, pin wrench
23	19	UV test duration to degrade rubber parts	5	hr	400+	250	400+	400+	400+	250
24	19	Monster cycles to failure	5	Cycles	500k+	500k+	500k+	480k	500k+	330k
25	20	Japan Industrial Standards test	5	Binary	Pass	Pass	Pass	Pass	Pass	Pass
26	20	Bending strength (frontal loading)	5	kN	5.5	8.9	7.5	7.5	6.2	10.2

EXHIBIT 6-6 *Continued*

The desirable range of values for one metric may depend on another. In other words, we may wish to express a target as, for example, "the fork tip lateral stiffness is no more than 20 percent of the lateral stiffness at the brake pivots." In applications where the team feels this level of complexity is warranted, such targets can easily be included, although we recommend that this level of complexity not be introduced until the final phase of the specifications process.

Using these five different types of expressions for values of the metrics, the team sets the target specifications. The team simply proceeds down the list of metrics and assigns both the marginally acceptable and ideal target values for each metric. These decisions are facilitated by the metric-based competitive benchmarking chart shown in Exhibit 6-6. To set the target values, the team has many considerations, including the capability of competing products available at the time, competitors' future product capabilities (if these are predictable), and the product's mission statement and target market segment. Exhibit 6-8 shows the targets assigned for the suspension fork.

No.	Need	Imp.	ST Tritrack	Manray 2	Rox Tahx Quadra	Rox Tahx TI 21	Tonka Pro	Gunhill Head Shox
1	Reduces vibration to the hands	3	.		..		..	...
2	Allows easy traversal of slow, difficult terrain	2	..		...		...	
3	Enables high-speed descents on bumpy trails	5	.		..		..	...
4	Allows sensitivity adjustment	3	.		..		..	...
5	Preserves the steering characteristics of the bike	4		..	.	..		
6	Remains rigid during hard cornering	4	.	...	.		.	
7	Is lightweight	4	.	...	.	...		
8	Provides stiff mounting points for the brakes	2	.		...	...		..
9	Fits a wide variety of bikes, wheels, and tires	5			...		...	.
10	Is easy to install	1						.
11	Works with fenders	1	...	.	.	.	.	
12	Instills pride	5	.		...		...	
13	Is affordable for an amateur enthusiast	5		.	...	.	...	..
14	Is not contaminated by water	5	.	...			..	
15	Is not contaminated by grunge	5	.	...	.		..	
16	Can be easily accessed for maintenance	3						.
17	Allows easy replacement of worn parts	1						.
18	Can be maintained with readily available tools	3					..	.
19	Lasts a long time	5				...		.
20	Is safe in a crash	5						

EXHIBIT 6-7 Competitive benchmarking chart based on perceived satisfaction of needs. (Scoring more “dots” corresponds to greater perceived satisfaction of the need.)

Because most of the values are expressed in terms of bounds (upper or lower or both), the team is establishing the boundaries of the competitively viable product space. The team hopes that the product will meet some of the ideal targets but is confident that a product can be commercially viable even if it exhibits one or more marginally acceptable characteristics. Note that these specifications are preliminary because until a product concept is chosen and some of the design details are worked out, many of the exact trade-offs are uncertain.

Step 4: Reflect on the Results and the Process

The team may require some iteration to agree on the targets. Reflection after each iteration helps to ensure that the results are consistent with the goals of the project. Questions to consider include:

- Are members of the team “gaming”? For example, is the key marketing representative insisting that an aggressive value is required for a particular metric in the hopes that by setting a high goal, the team will actually achieve more than if his or her true, and more lenient, beliefs were expressed?
- Should the team consider offering multiple products or at least multiple options for the product to best match the particular needs of more than one market segment, or will one “average” product suffice?
- Are any specifications missing? Do the specifications reflect the characteristics that will dictate commercial success?

Once the targets have been set, the team can proceed to generate solution concepts. The target specifications then can be used to help the team select a concept and will help the team know when a concept is commercially viable. (See Chapter 7, Concept Generation, and Chapter 8, Concept Selection.)

Setting the Final Specifications

As the team finalizes the choice of a concept and prepares for subsequent design and development, the specifications are revisited. Specifications that originally were only targets expressed as broad ranges of values are now refined and made more precise.

Finalizing the specifications is difficult because of trade-offs—inverse relationships between two specifications that are inherent in the selected product concept. Trade-offs frequently occur between different technical performance metrics and almost always occur between technical performance metrics and production cost. There may also be trade-offs between product performance and development time or cost. For example, one trade-off is between brake mounting stiffness and mass of the fork. Because of the basic mechanics of the fork structure, these specifications are inversely related, assuming other factors are held constant. Another trade-off is between cost and mass. For a given concept, the team may be able to reduce the mass of the fork by making some parts out of titanium instead of steel. Unfortunately, decreasing the mass in this way will most likely increase the manufacturing cost of the product. The difficult part of refining the specifications is choosing how such trade-offs will be resolved.

begins. Ideally the team has been involved both in the identification of the customer needs and in the setting of the target product specifications. Those members of the team who were not involved in these preceding steps should become familiar with the processes used and their results before concept generation activities begin. (See Chapter 5, Identifying Customer Needs, and Chapter 6, Product Specifications.)

As stated before, the challenge was to “design a better handheld roofing nailer.” The scope of the design problem could have been defined more generally (e.g., “fasten roofing materials”) or more specifically (e.g., “improve the speed of the existing pneumatic tool concept”). Some of the assumptions in the team’s mission statement were:

- The nailer will use nails (as opposed to adhesives, screws, etc.).
- The nailer will be compatible with nail magazines on existing tools.
- The nailer will nail through roofing shingles into wood.
- The nailer will be handheld.

Based on the assumptions, the team had identified the customer needs for a handheld nailer. These included:

- The nailer inserts nails in rapid succession.
- The nailer is lightweight.
- The nailer has no noticeable nailing delay after tripping the tool.

The team gathered supplemental information to clarify and quantify the needs, such as the approximate energy and speed of the nailing. These basic needs were subsequently translated into target product specifications. The target specifications included the following:

- Nail lengths from 25 millimeters to 38 millimeters.
- Maximum nailing energy of 40 joules per nail.
- Nailing forces of up to 2,000 newtons.
- Peak nailing rate of one nail per second.
- Average nailing rate of 12 nails per minute.
- Tool mass less than 4 kilograms.
- Maximum trigger delay of 0.25 second.

Decompose a Complex Problem into Simpler Subproblems

Many design challenges are too complex to solve as a single problem and can be usefully divided into several simpler subproblems. For example, the design of a complex product like a document copier can be thought of as a collection of more focused design problems, including, for example, the design of a document handler, the design of a paper feeder, the design of a printing device, and the design of an image capture device. In some cases, however, the design problem cannot readily be divided into subproblems. For example, the problem of designing a paper clip may be hard to divide into subproblems. As a general rule, we feel that teams should attempt to decompose design problems, but should be aware that such a decomposition may not be very useful for products with extremely simple functions.

Dividing a problem into simpler subproblems is called *problem decomposition*. There are many schemes by which a problem can be decomposed. Here we demonstrate a *functional* decomposition and also list several other approaches that are frequently useful.

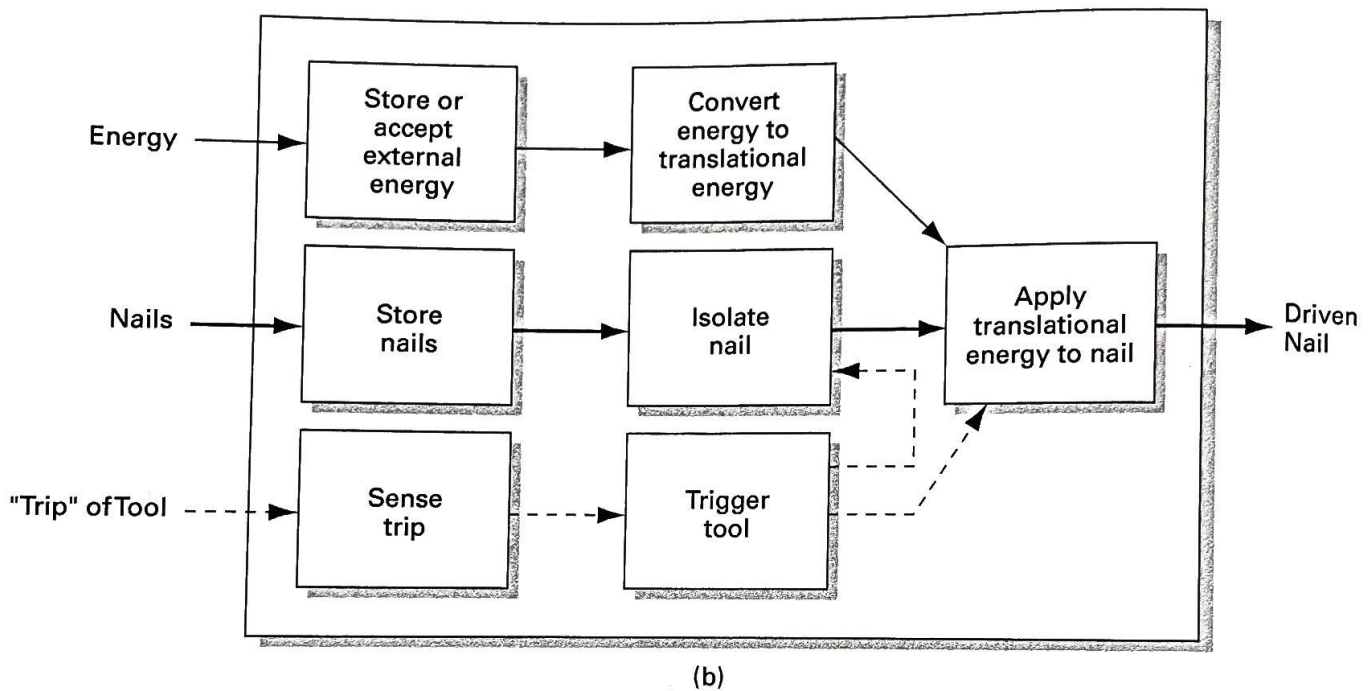
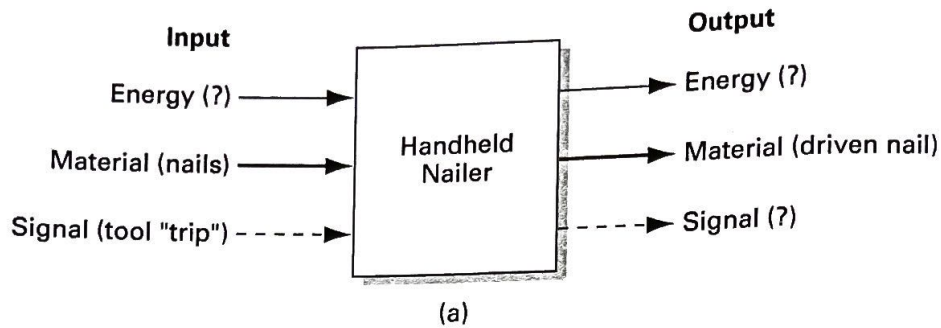


EXHIBIT 7-4 Function diagram of a handheld nailer arising from a functional decomposition: (a) overall "black box"; (b) refinement showing subfunctions.

The first step in decomposing a problem functionally is to represent it as a single *black box* operating on material, energy, and signal flows, as shown in Exhibit 7-4(a). Thin solid lines denote the transfer and conversion of energy, thick solid lines signify the movement of material within the system, and dashed lines represent the flows of control and feedback signals within the system. This black box represents the overall function of the product.

The next step in functional decomposition is to divide the single black box into subfunctions to create a more specific description of what the elements of the product might do to implement the overall function of the product. Each subfunction can generally be further divided into even simpler subfunctions. The division process is repeated until the team members agree that each subfunction is simple enough to work with. A good rule of thumb is to create between 3 and 10 subfunctions in the diagram. The end result, shown in Exhibit 7-4(b), is a function diagram containing subfunctions connected by energy, material, and signal flows.

Note that at this stage the goal is to describe the functional elements of the product without implying a specific technological working principle for the product concept. For example, Exhibit 7-4(b) includes the subfunction "isolate nail." This subfunction is expressed in such a way that it does not imply any particular physical solution concept, such as indexing the coil of nails into a slot or breaking a nail sideways off of the stick. The team should consider each subfunction in turn and ask whether it is expressed in a way that does not imply a particular physical solution principle.

There is no single correct way of creating a function diagram and no single correct functional decomposition of a product. A helpful way to create the diagram is to quickly create several drafts and then work to refine them into a single diagram that the team is comfortable with. Some useful techniques for getting started are:

- Create a function diagram of an existing product.
- Create a function diagram based on an arbitrary product concept already generated by the team or based on a known subfunction technology. Be sure to generalize the diagram to the appropriate level of abstraction.
- Follow one of the flows (e.g., material) and determine what operations are required. The details of the other flows can be derived by thinking about their connections to the initial flow.

Note that the function diagram is typically not unique. In particular, subfunctions can often be ordered in different ways to produce different function diagrams. Also note that in some applications the material, energy, and signal flows are difficult to identify. In these cases, a simple list of the subfunctions of the product, without connections between them, is often sufficient.

Functional decomposition is most applicable to technical products, but it can also be applied to simple and apparently nontechnical products. For example, an ice cream scoop has material flow of ice cream being separated, formed, transported, and deposited. These subfunctions could form the basis of a problem decomposition.

Functional decomposition is only one of several possible ways to divide a problem into simpler subproblems. Two other approaches are:

- **Decomposition by sequence of user actions:** For example, the nailer problem might be broken down into three user actions: moving the tool to the gross nailing position, positioning the tool precisely, and triggering the tool. This approach is often useful for products with very simple technical functions involving a lot of user interaction.
- **Decomposition by key customer needs:** For the nailer, this decomposition might include the following subproblems: fires nails in rapid succession, is lightweight, and has a large nail capacity. This approach is often useful for products in which form, and not working principles or technology, is the primary problem. Examples of such products include toothbrushes (assuming the basic brush concept is retained) and storage containers.

Focus Initial Efforts on the Critical Subproblems

The goal of all of these decomposition techniques is to divide a complex problem into simpler problems such that these simpler problems can be tackled in a focused way. Once problem decomposition is complete, the team chooses the subproblems that are most critical to the success of the product and that are most likely to benefit from novel or creative